



EXPERIMENT 1.1

Getting to know your microscope

Aim

To experiment with a compound light microscope.

Materials

Compound light microscope
Microscope slide
Coverslip
Small piece of newspaper
Small piece of tissue paper
Hair (use your own)
1 cm sticky tape (transparent)
Eyedropper
Small beaker of water

Method

- 1 Cut out two small words from a piece of newspaper.
- 2 Place the cut out newspaper on the microscope slide and add two drops of water to help it 'stick' to the slide. Place a coverslip on top. This is called a **wet mount**.
- 3 Follow the instructions for using a microscope on page 10. On the lowest magnification, find one letter from the newsprint to focus on.
- 4 Move the slide slightly towards your body and observe what happens.
- 5 Move the slide slightly to the left and observe what happens.
- 6 Change the magnification and observe what happens.
- 7 Draw a diagram of the newspaper letter.

Take the newspaper out and prepare another slide using the tissue paper. Make sure the drop of water is added and the coverslip is placed over the top carefully.

- 9 Sketch what you see.
- 10 Repeat with sticky tape and then a hair from your head.

Results

Include your diagrams here.

Discussion

- 1 Describe what the newspaper letter looks like through the microscope. What does this mean for all things you see through this type of microscope?

Cause

What did you do to cause the change you observed?



Effect

What effect did it have?

→ Fig 1.18 Cause-and-effect graphic organiser.

- 2 What features could you see on the tissue paper and sticky tape that you could not see with the naked eye?
- 3 Use a series of cause-and-effect graphic organisers, similar to that shown in Figure 1.18, to record the results of your experiment when you moved the slide different ways. For example, the cause link may be 'move the slide to the left'. Then write what happened in the effect link.

Conclusion

What do you know about compound light microscopes?

